Date

LECTURE-12 | Monday 12/10/26.

Control Hazards: Related to branching / jump

- Most critical since we need to know PC. Agar jump ni lagna ho but hamnay laga dia ho tou ROLLBack / Instr. flush krna hoga tou time lagay ga.
- Jump done = Taken (T)
  Jump not done = Not Taken (NT)

- Therefore we try to predict in hardware level in ALU so we can calculate future's PC to know wheather T/NT branch.

- We can predict on software level too which is either: static or dynamic.
  Usually forward jump is declared taken & backward not taken as forward jump is PC+4 is easier than PC - offset.
- If predicted value is wrong (jump to be taken but we didn't take), then we'll do roll back & flush all instrc. since beginning of that branch (git reset-hard).

Pipelining registers: These Registers are blw every stage. IF/ID IF ↑ ID

NOTE: dest. & source reg. ≠ 0 as 0x0 is reserved for zero value.

This provides forwarding which is more baificial as compared to overhead time to read & write from reg.

Date

fix

⇒ Static Prediction : We either do all "Taken" or all "Not Taken"

Q:

Actual Results: NT, T, T, T, NT, T, NT, T, NT, T

Prediction Static: T, T, T, T, T, T, T, T, T, T

Result ✗ ✓ ✓ ✓ ✗ ✓ ✗ ✓ ✗ ✓

6/10 = ✓ ∴ 60% Accurate prediction